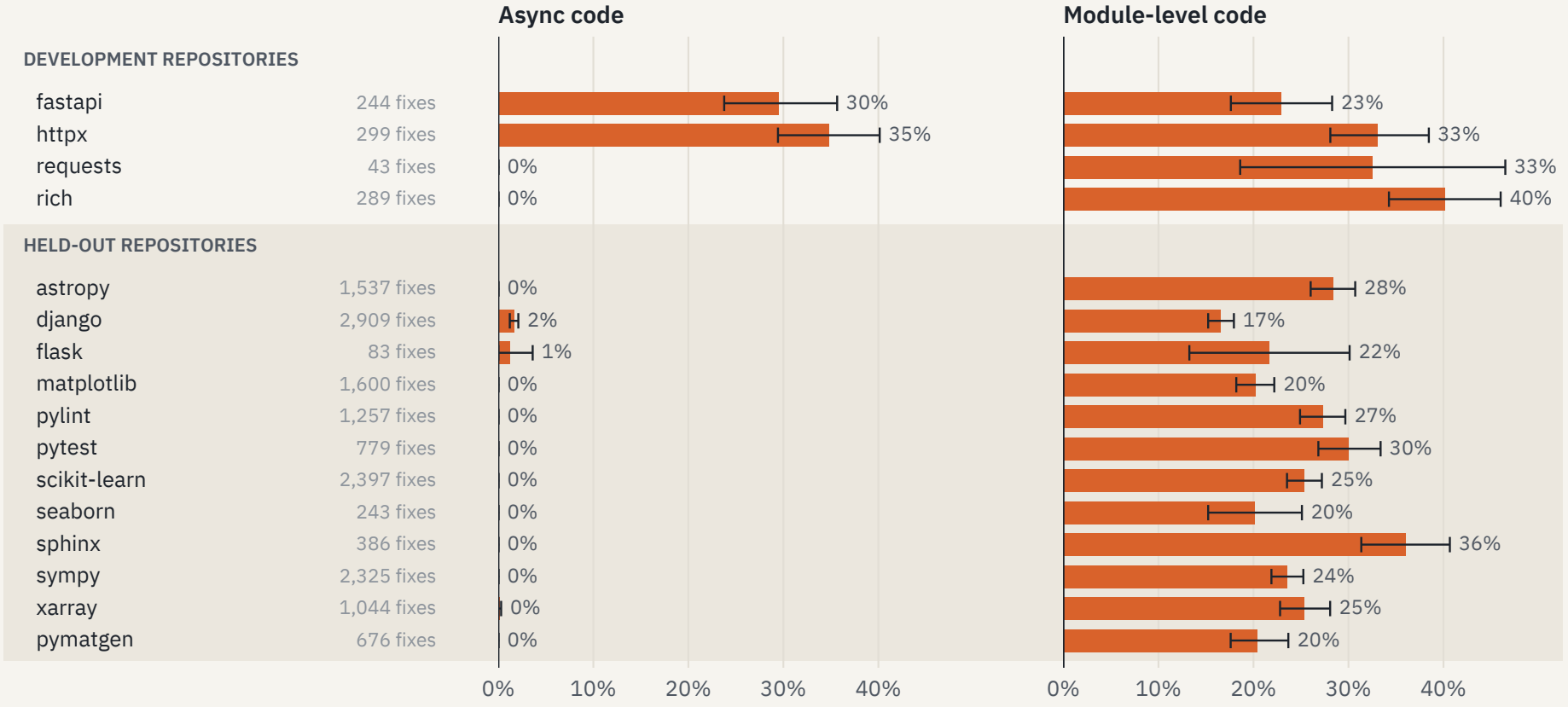
Fixes touch async code in two libraries, module-level code everywhere

Share of code+test fixes that touch each kind of code, per repository; whiskers: 95% bootstrap CI



Source: cgl-profile on each repository's history, 2019 to Sep 2026 (pymatgen: to Mar 2026). Module-level code includes new top-level definitions.